Session 9A2: Graph Drawing Toolkit — How to Draw Any Function

Phase 2 — Classical Techniques | 75 min

No calculus required. Just pencil, paper, and a systematic mind.

Part A: The 7-Step Drawing Sequence

> Every graph can be drawn by answering seven questions in order. Memorize this sequence.

```
Step 1 — Domain: Which x-values can I draw?
Step 2 — Symmetry: Can I draw only half?
Step 3 — Intercepts: Where does it cross the axes?
Step 4 — Asymptotes: What lines does it approach but never touch?
Step 5 — Sign: Above or below the x-axis?
Step 6 — Connect: Smooth curve through the dots.
Step 7 — Special: Holes? Steps? Sawteeth? Handle separately.
```

Example 1: Domain — The Four Rules of Entry

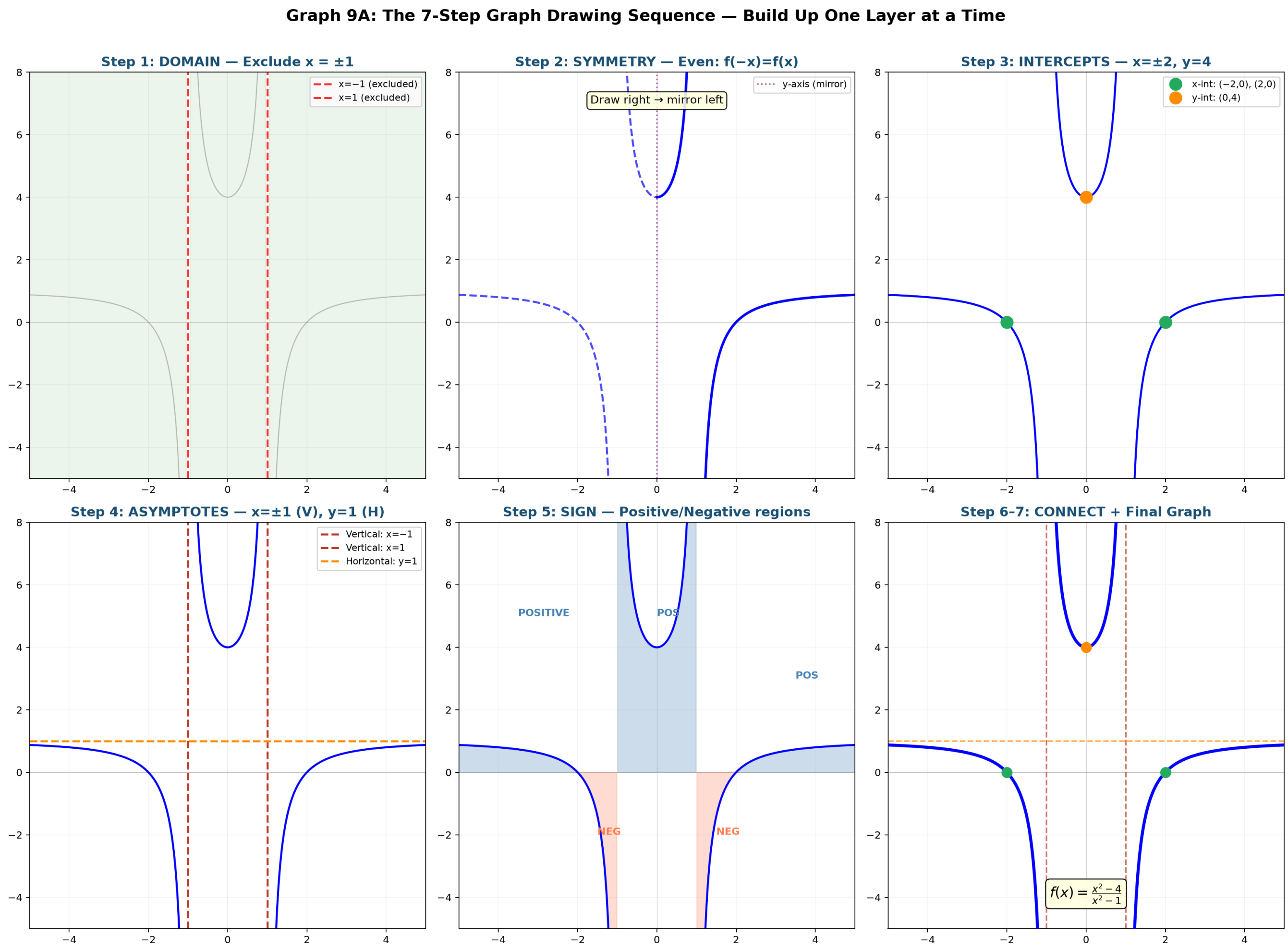
Rule 1 — Square Root: $\sqrt{x-2}$.
Inside must be ≥ 0. $x - 2 \geq 0 \to x \geq 2$.

Rule 2 — Denominator: $\frac{1}{x^2-4}$.
Denominator $\neq 0$. $x^2 - 4 = 0 \to x \neq 2, -2$.

Rule 3 — Both Mixed: $\frac{\sqrt{x+1}}{x-2}$.
$\sqrt{\cdot} \to x \geq -1$. Denominator $\to x \neq 2$. Domain: $[-1, 2) \cup (2, \infty)$.

Rule 4 — Logarithm: $\log_3(2x-5)$.

Argument > 0. $2x - 5 > 0 \to x > \frac{5}{2}$.



Graph 9A: The 7-step method in action — Domain → Symmetry → Intercepts → Asymptotes → Sign → Connect. Each panel adds one layer of understanding until the full graph emerges.

Example 2: Polynomial — Symmetry, Sign Chart, End Behavior

$f(x) = x^3 - 4x$.

Step 1 — Domain: All real numbers.

Step 2 — Symmetry: $f(-x) = -x^3 + 4x = -(x^3 - 4x) = -f(x)$. → **Odd**, origin symmetry. Draw right side, spin 180°.

Step 3 — Intercepts: $x^3 - 4x = x(x-2)(x+2) = 0$. Roots: $-2, 0, 2$.

Step 5 — Sign chart: Roots split into four intervals.

$x < -2$: all factors negative → **below**.

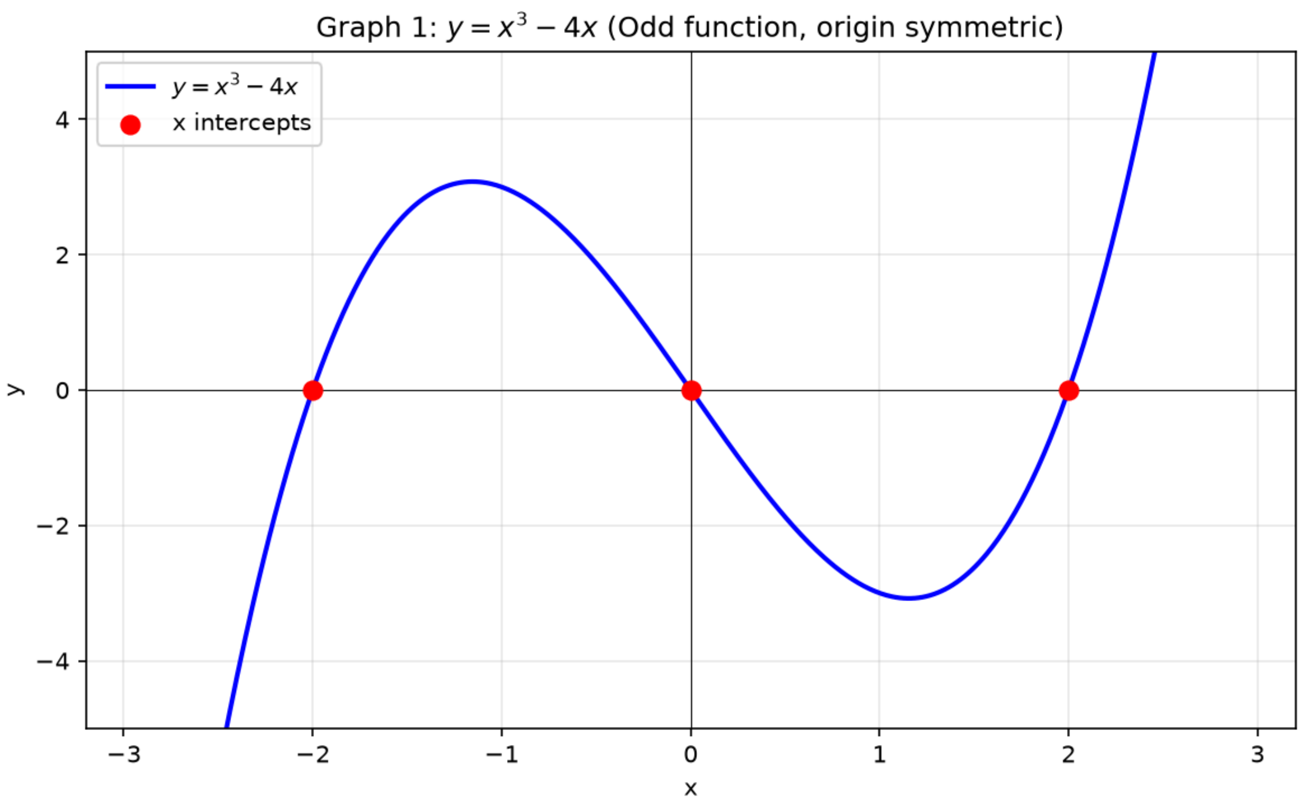
$-2 < x < 0$: two negatives → **above**.

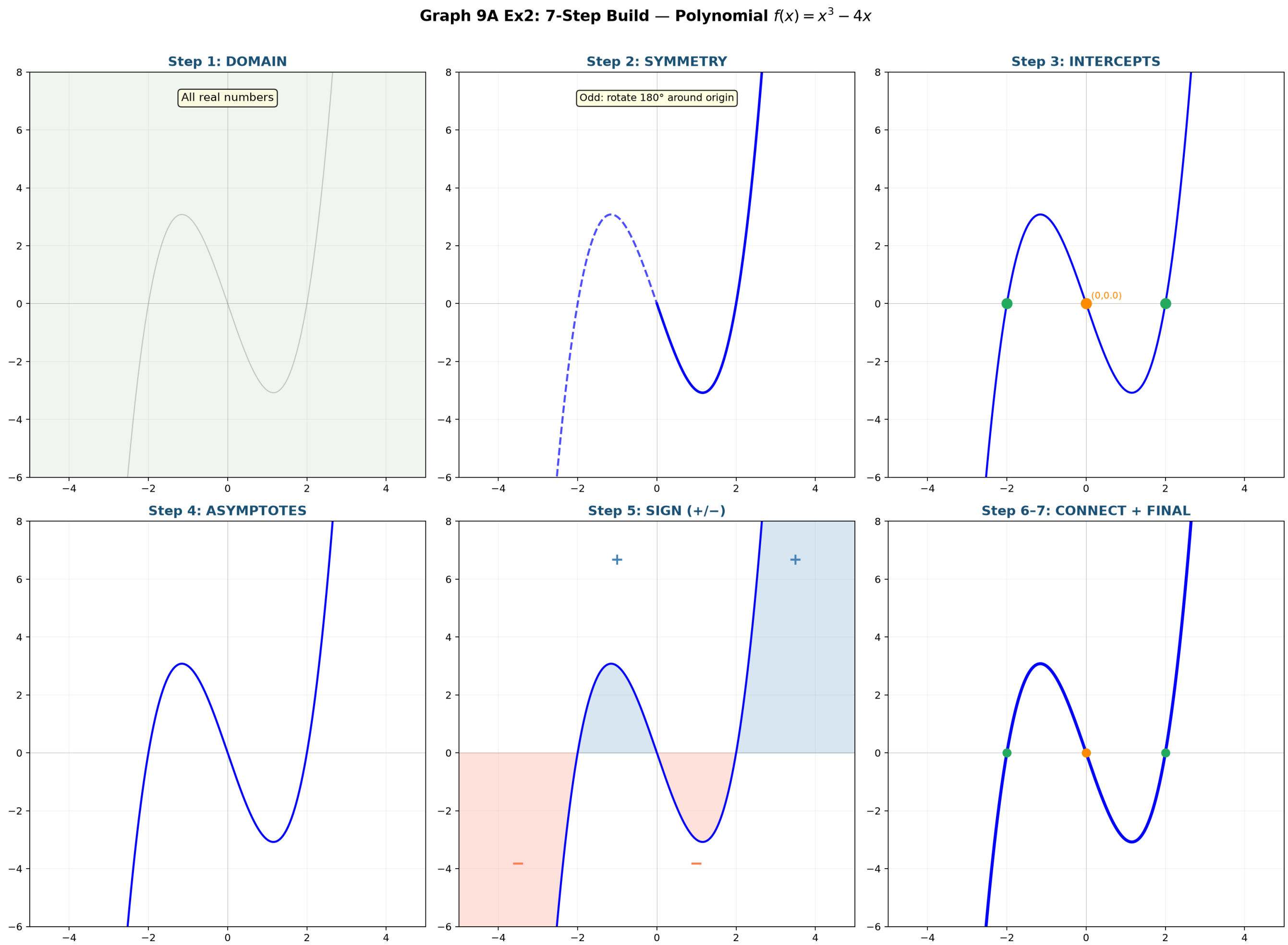
$0 < x < 2$: one negative → **below**.

$x > 2$: all positive → **above**.

End behavior: $x \to \infty$, x^3 dominates → $+\infty$. $x \to -\infty$ → $-\infty$.

Step 6 — Connect: From bottom left, cross up at -2, down at 0, up at 2, to top right.





Graph 9A Ex2: The 7-step method applied to a cubic — odd symmetry halves the work, the sign chart reveals four alternating regions, and end behavior sends the arms to ±∞.

Example 3: Rational Function — Tear, Cancel, Punch a Hole

$f(x) = \frac{x^2-x-2}{x^2-4}$.

Step 0 — Factor: Numerator $(x-2)(x+1)$. Denominator $(x-2)(x+2)$. Cancel $(x-2)$. → $f(x) = \frac{x+1}{x+2}$, **but** $x \neq 2$.

$x = 2$ made the denominator 0. It canceled but is forever banned.
→ **Empty circle (hole)** at $(2, \frac{3}{4})$.

Step 1 — Domain: $x \neq -2$ (zero denominator), $x \neq 2$ (hole).

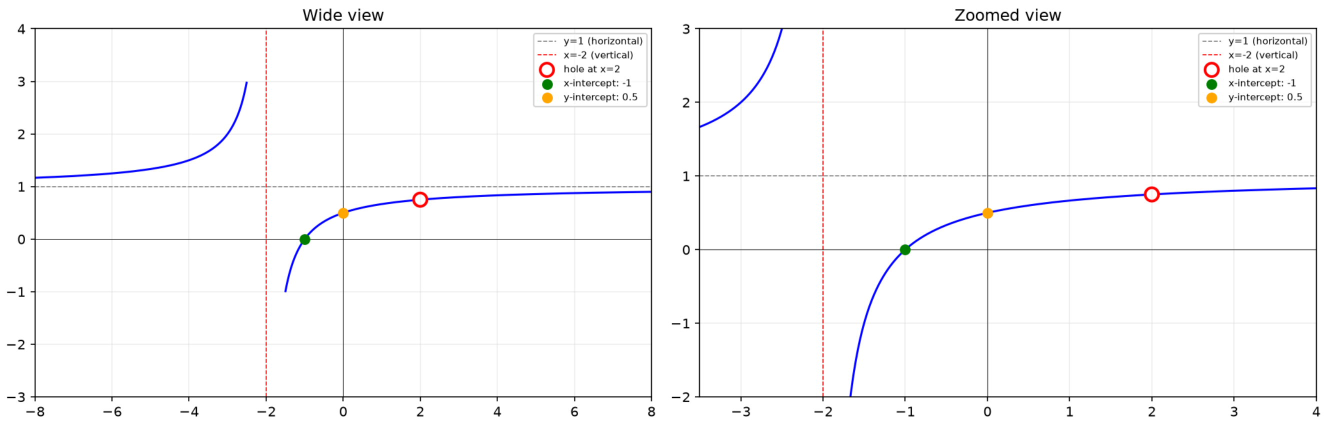
Step 4 — Asymptotes: Vertical $x = -2$. Horizontal as $x \to \infty$: ratio of leading coefficients → $y = 1$.

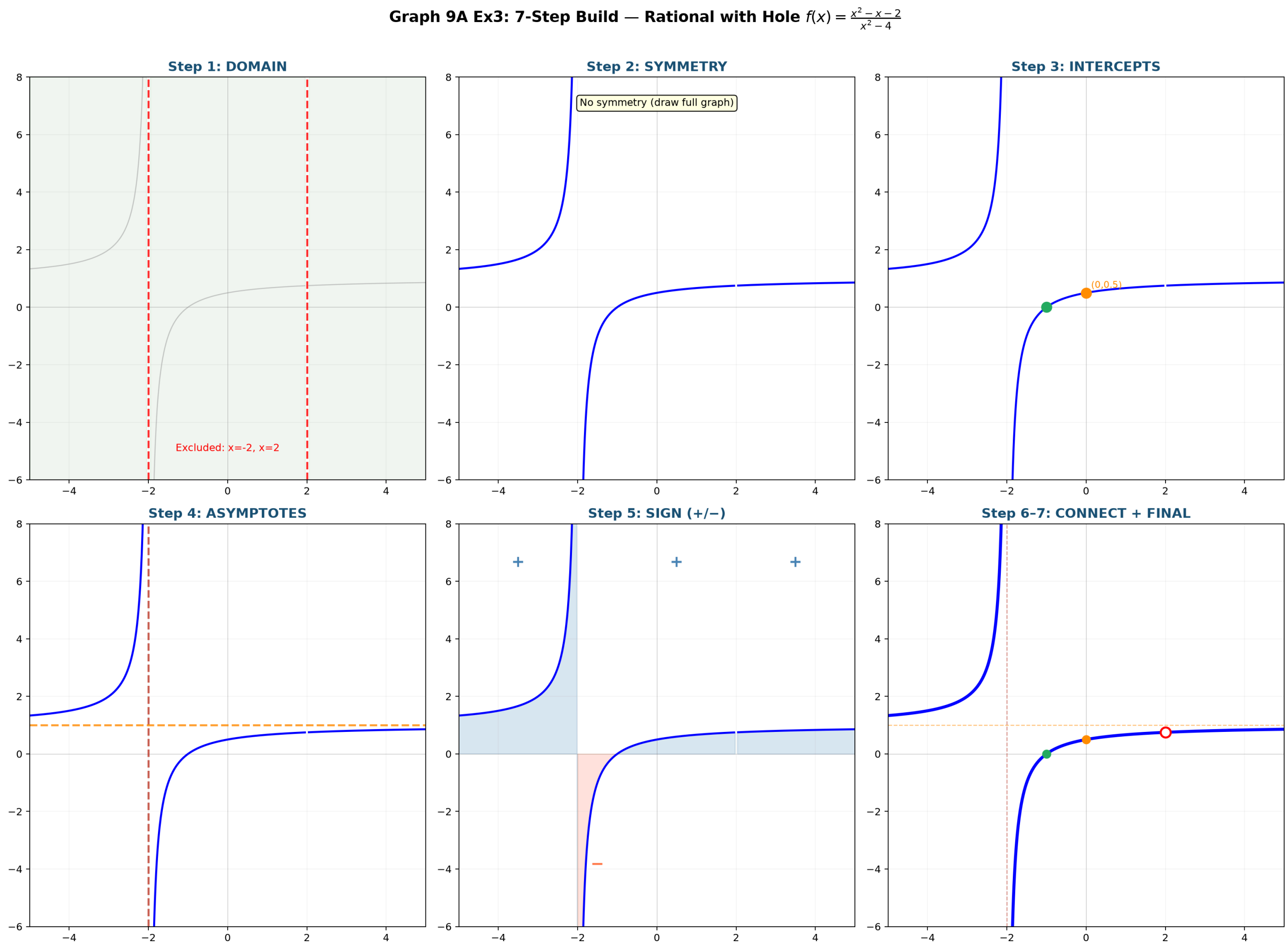
Step 3 — Intercepts: y-int: $f(0) = \frac{1}{2}$. x-int: numerator $= 0$ at $x = -1$.

Step 5 — Sign near $x = -2$:
$x \to -2^{+}$: numerator (-1), denominator $(+)$ → $-\infty$.
$x \to -2^{-}$: numerator (-1), denominator $(-)$ → $+\infty$.

Step 6 — Connect: Two branches, hole at $x = 2$.

Graph 2: $y = \frac{x^2 - x - 2}{x^2 - 4}$ (hole at x=2)





Graph 9A Ex3: Factor → cancel → hole at x=2. The 7-step method catches the hole (Step 7), the vertical asymptote at x=−2, and the horizontal asymptote y=1.

Example 4: Slant Asymptote — Divide First

$f(x) = \frac{x^2+2x}{x-1}$.

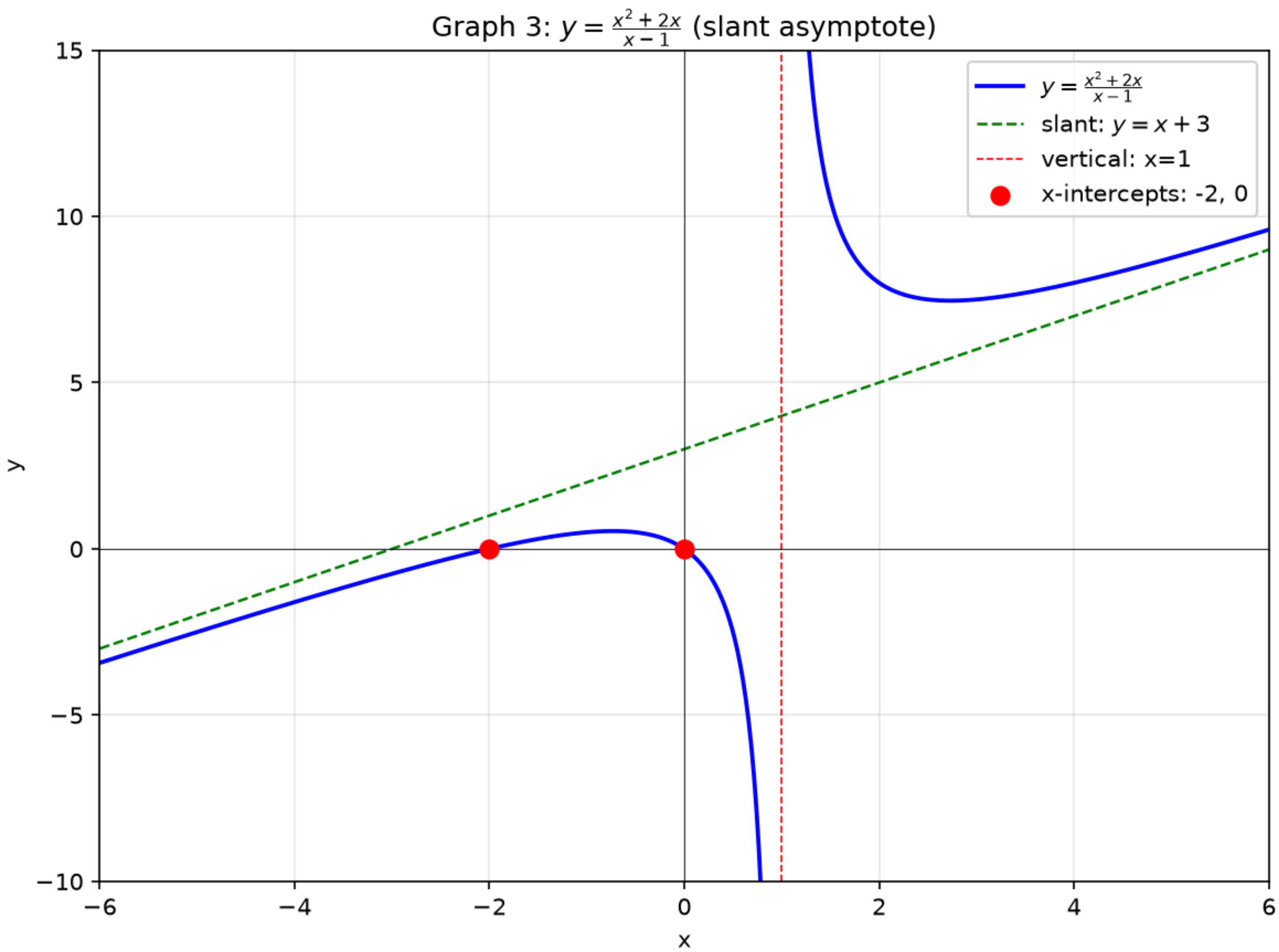
Divide: $x^2 + 2x \div (x - 1) = x + 3$ with remainder 3.
→ $f(x) = x + 3 + \frac{3}{x-1}$.

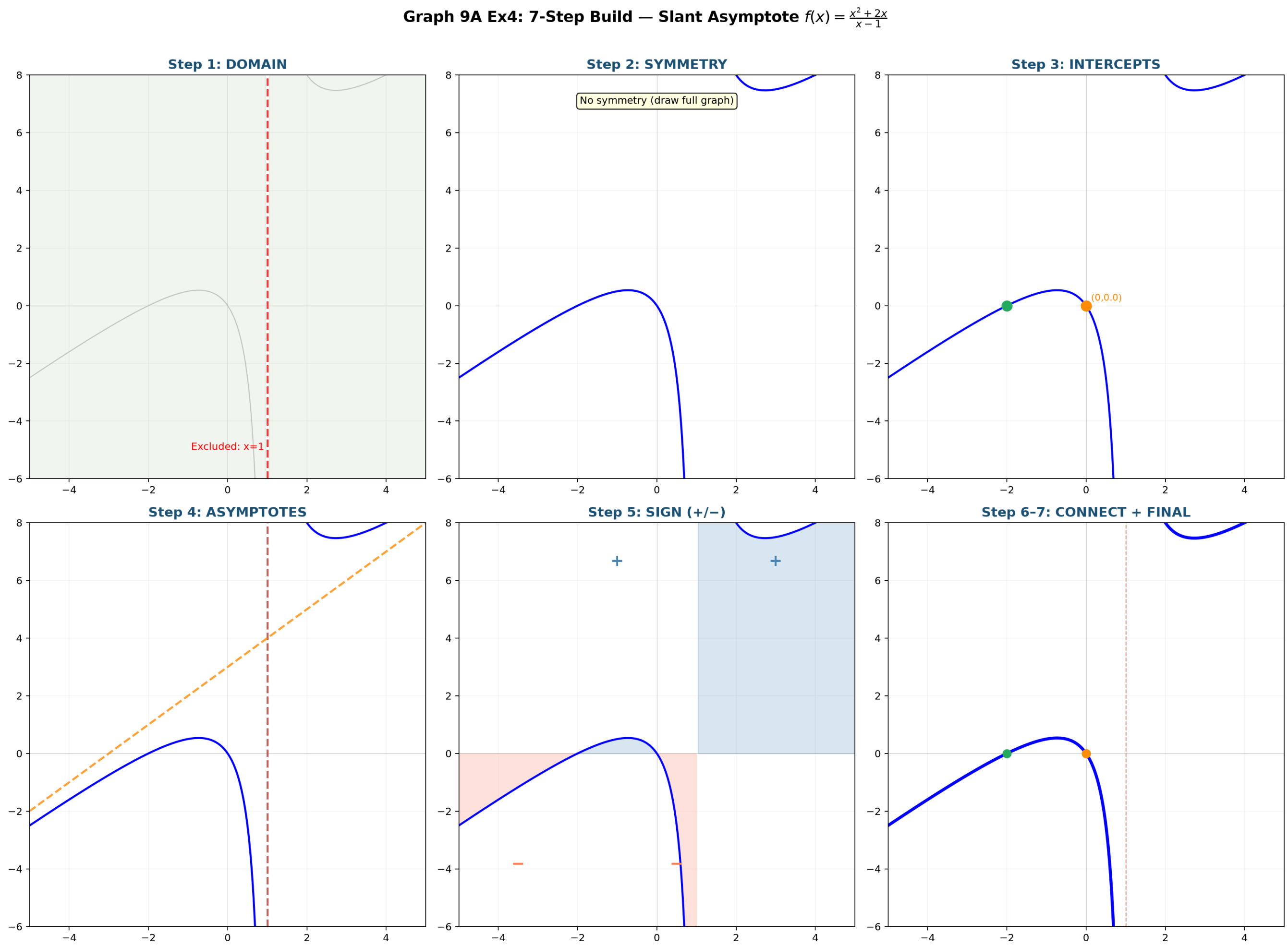
Vertical: $x = 1$. **Slant**: as $x \to \pm\infty$, $\frac{3}{x-1} \to 0$ → hugs $y = x + 3$.

Intercepts: $x = 0 \to f(0) = 0$. $f(x) = 0 \to x(x + 2) = 0 \to x = 0, -2$.

Drawing order: ① Dashed slant line $y = x + 3$. ② Dashed vertical $x = 1$. ③ Mark intercepts.

④ Curve hugging asymptotes.





Graph 9A Ex4: Step 4 reveals the slant asymptote y=x+3 (orange dashed) alongside the vertical x=1. The sign chart in Step 5 shows four alternating positive/negative regions.

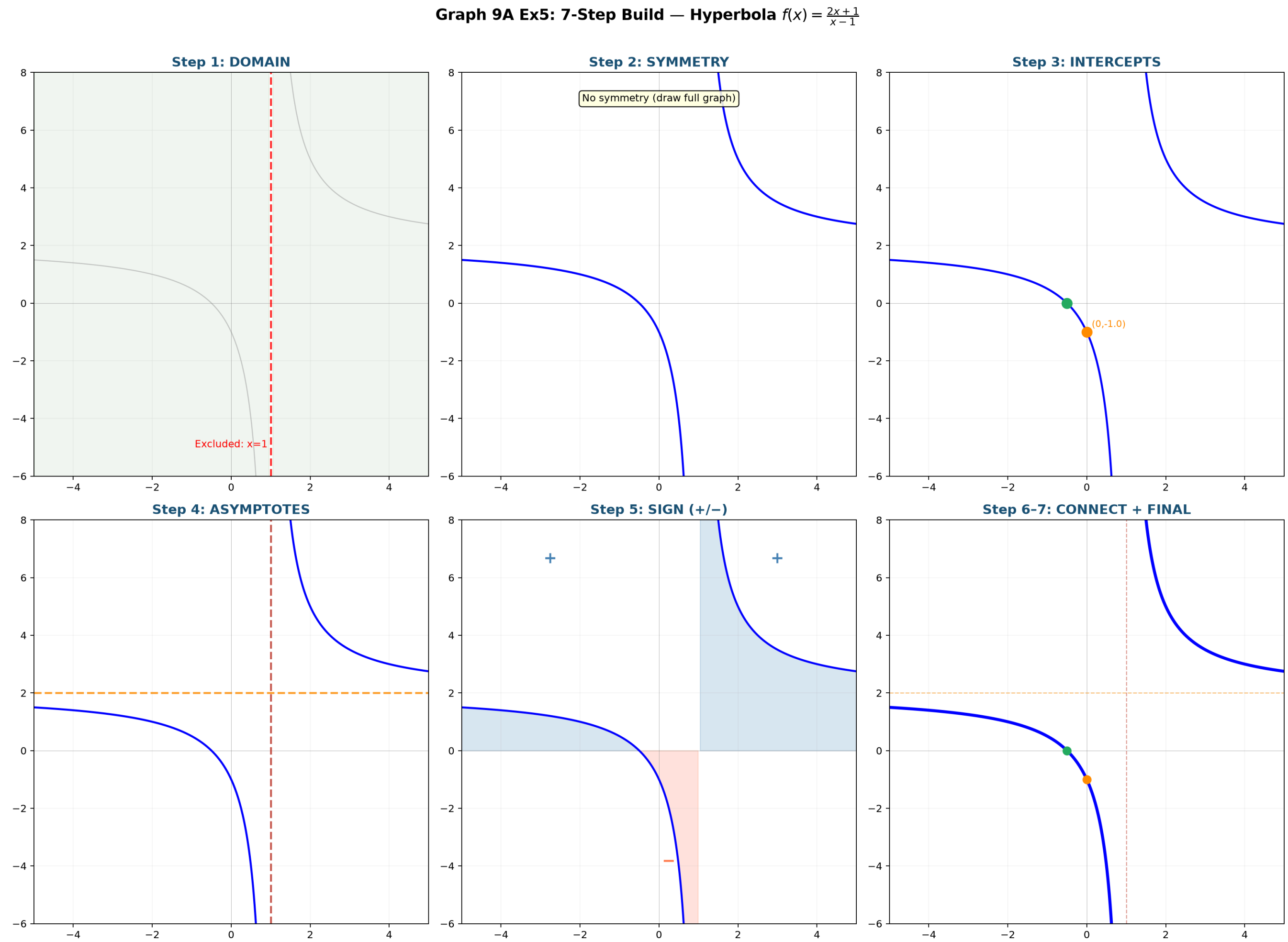
Example 5: The $\frac{ax+b}{cx+d}$ Hyperbola Form

$f(x) = \frac{2x+1}{x-1}$.

Vertical: $x = 1$. **Horizontal**: $y = 2$ (ratio of x-coefficients).

Intercepts: y-int $(0, -1)$. x-int $(-\frac{1}{2}, 0)$.

Center: intersection of asymptotes $(1, 2)$ — the graph is symmetric around this point.



Graph 9A Ex5: The hyperbola form (ax+b)/(cx+d). Steps 3–4 reveal intercepts and both asymptotes. The sign chart confirms the two branches live in opposite quadrants.

Approach: $x \to 1^+ \to +\infty$, $x \to 1^- \to -\infty$.

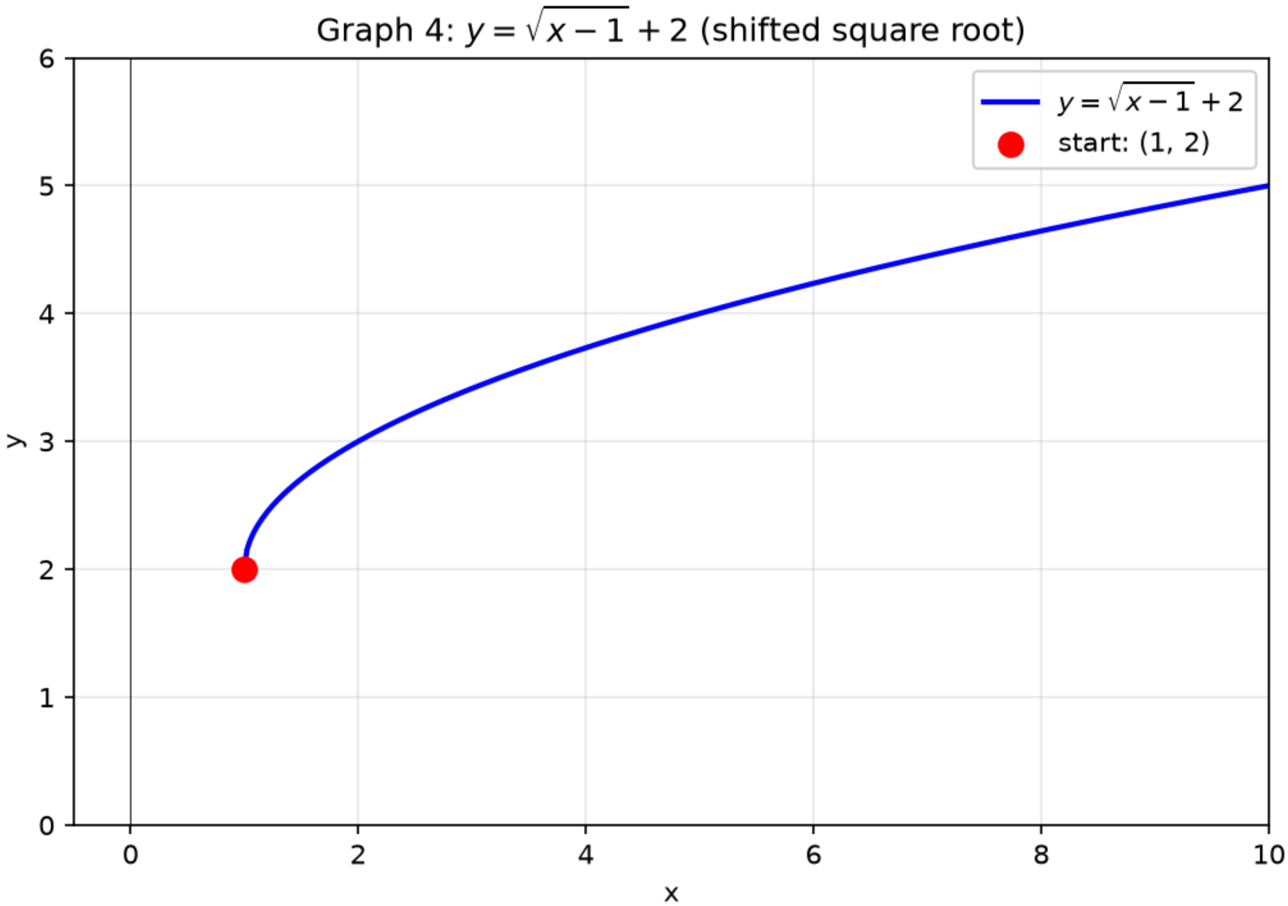
Example 6: Radical Function — The Half-Graph

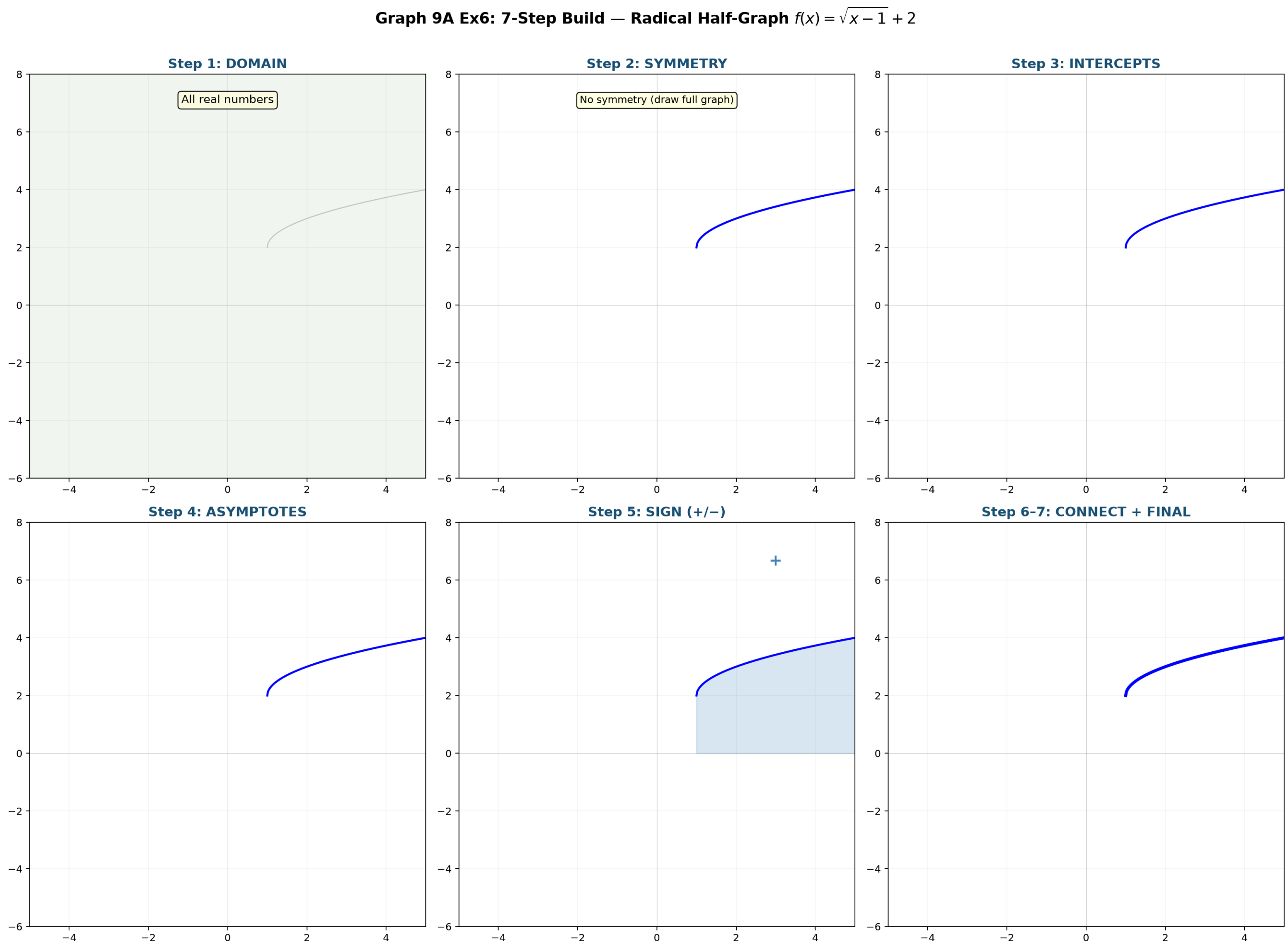
$f(x) = \sqrt{x-1} + 2$.

Starting point: $x = 1$ gives $f(1) = 2$. Mark $(1, 2)$.
For $x < 1$: nothing — the root is imaginary.

Growth: Slow. At $x = 5$: $\sqrt{4} + 2 = 4$. At $x = 17$: $\sqrt{16} + 2 = 6$.

Shape: $y = \sqrt{x}$ shifted right 1, up 2 — always creeping upward.


Graph 4: $y = \sqrt{x-1} + 2$ (shifted square root)
$y = \sqrt{x-1} + 2$
start: (1, 2)
x
y
0
2
4
6
8
10
1
3
5



Graph 9A Ex6: The radical function lives only where x≥1. Step 1 (domain) already tells you half the story — there is no left side. Step 5 (sign) confirms everything is positive.

> **Up to here**: 7-step sequence. Domain: 4 rules. Polynomial: symmetry + sign chart. Rational: cancel, hole, asymptotes. Slant: divide. Hyperbola: $\frac{ax+b}{cx+d}$. Radical: half-graph.

Part B: Transformations — Move, Flip, Fold, Stretch

Example 7: Move the Whole Graph

Base: $f(x) = |x|$ (V-shape, vertex at origin).

Operation	Formula	Vertex moves to
Right 3	$f(x-3) = \|x-3\|$	$(3, 0)$
Up 2	$f(x) + 2 = \|x\| + 2$	$(0, 2)$
Left 1, down 4	$f(x+1) - 4$	$(-1, -4)$

Method: draw the original, move the vertex, redraw the V.

Example 8: Flip Over an Axis

Base: $f(x) = \sqrt{x}$ (only $x \geq 0$, top-right quadrant).

Flip over x-axis: $y = -\sqrt{x}$. Every y becomes its negative. Points downward.
Flip over y-axis: $y = \sqrt{-x}$. Only $x \leq 0$. Mirror of the right side.
Flip over origin: $y = -\sqrt{-x}$. Quadrant 3 only. Both flips at once.

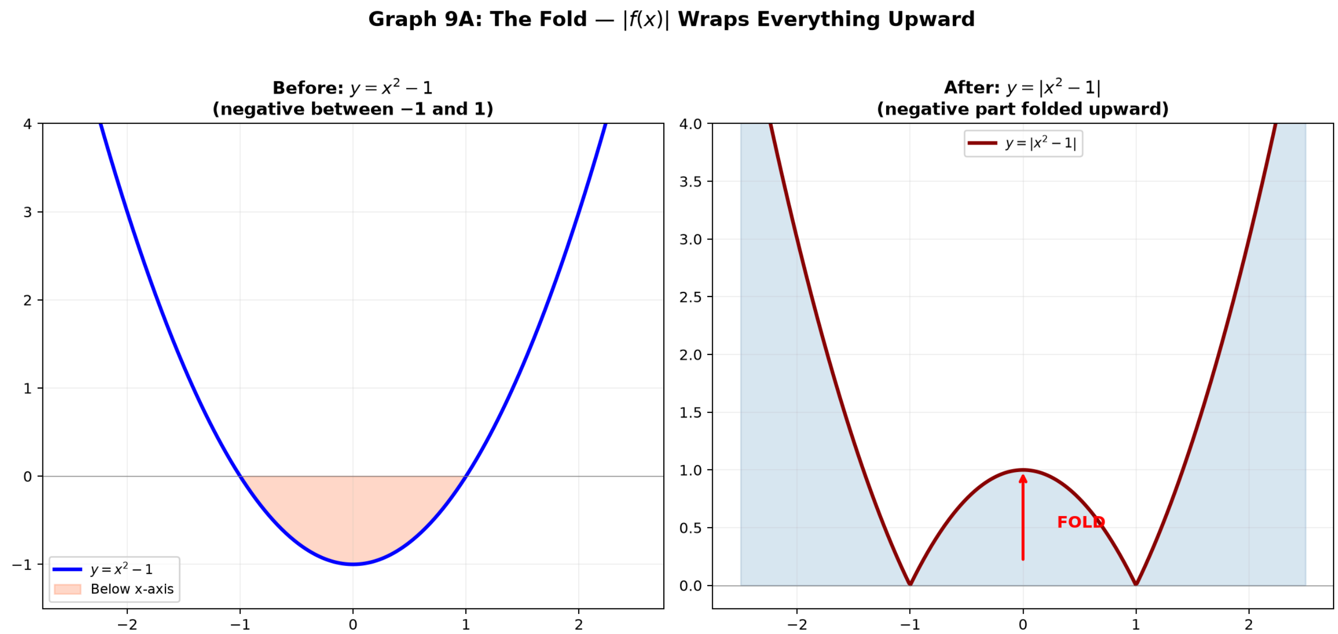
Example 9: Fold — Absolute Value Wraps Everything Upward

$y = |f(x)|$: Any part below the x-axis gets folded upward.

$f(x) = x^2 - 1$ dips to -1 between $-1 < x < 1$.
$|x^2 - 1|$: the dip becomes an upward bump. Everything is ≥ 0.

$y = f(|x|)$: Copy the right side ($x \geq 0$) onto the left.

$f(x) = x^2 - 2x$. For $x \geq 0$: original. For $x < 0$: reflect right side across y-axis.
Result: W-shape symmetric about y-axis.



Graph 9A: Left — f(x)=x²–1 dips below the axis. Right — |f(x)| folds the negative part upward, creating a W-shape between –1 and 1.

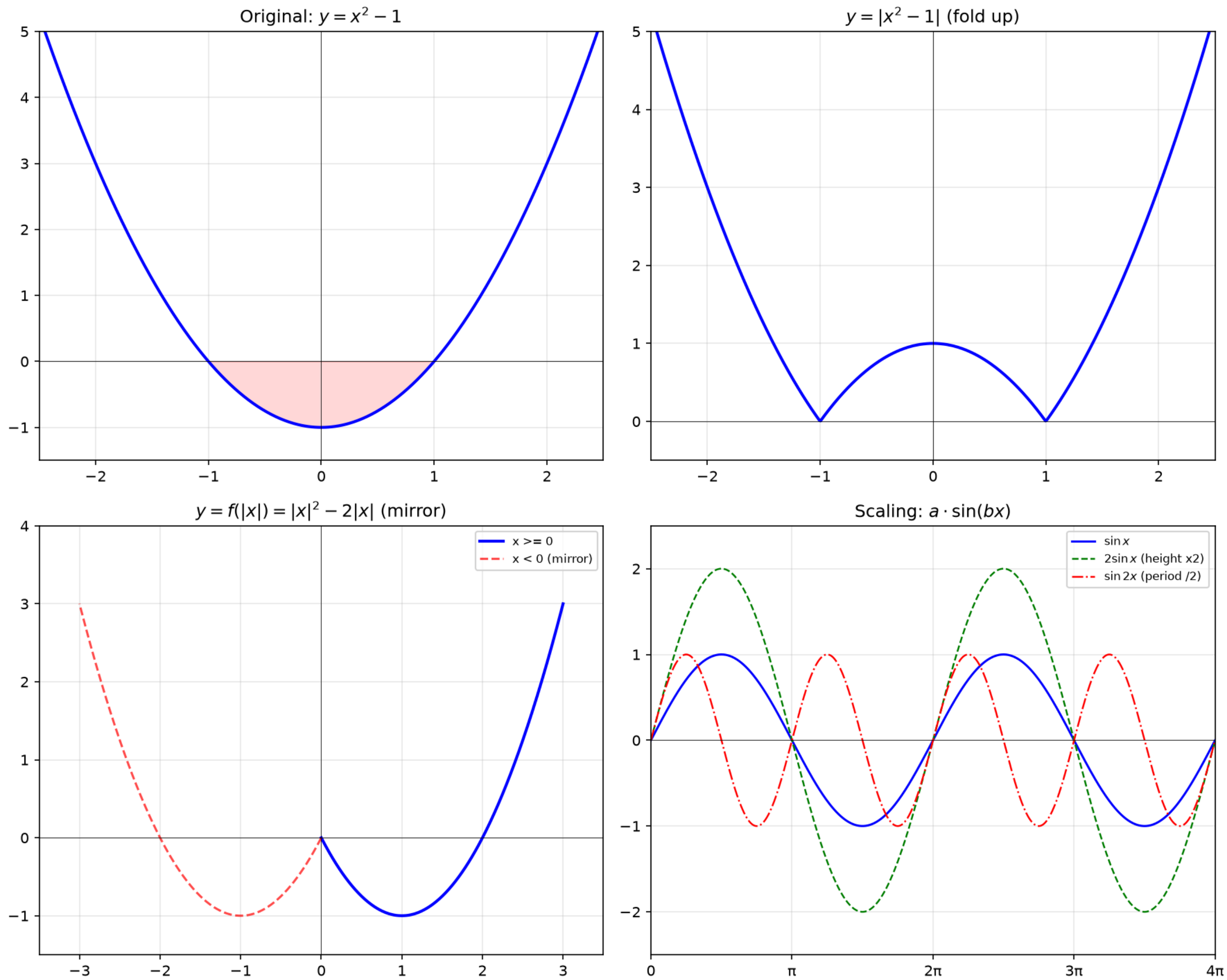
Example 10: Stretch and Shrink

Base: $f(x) = \sin x$ (height 1, period 2π).

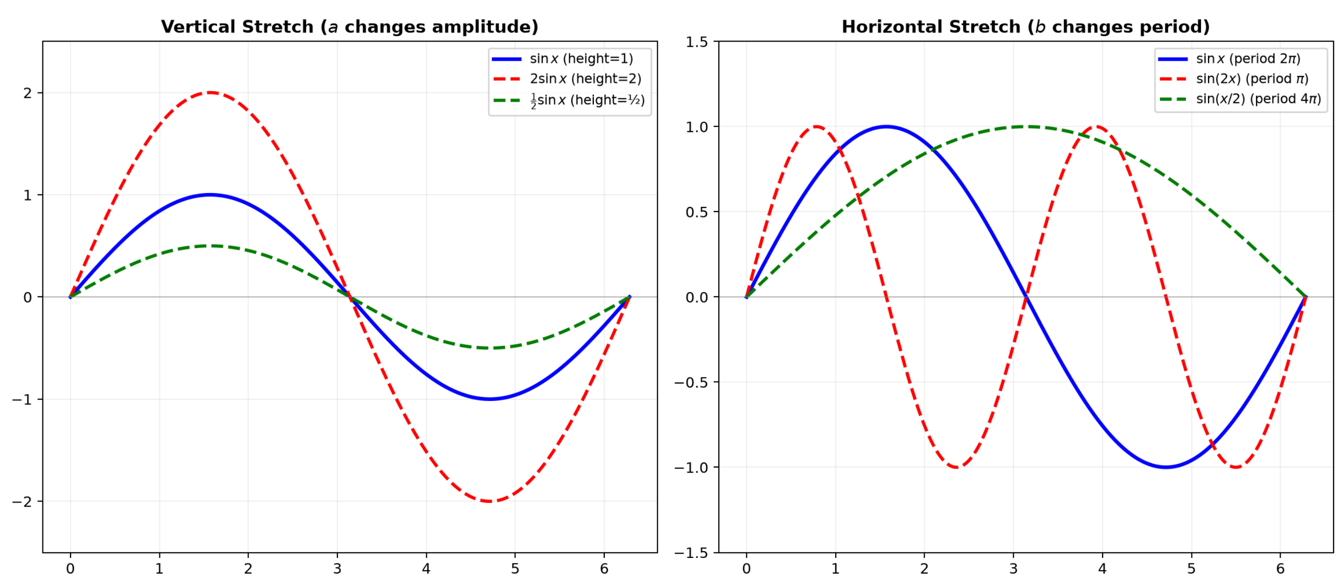
Operation	Effect
$2\sin x$	Height ×2 (vertical stretch)
$\frac{1}{2}\sin x$	Height ÷2 (vertical shrink)
$\sin(2x)$	Period ÷2 → π (horizontal squeeze)
$\sin(\frac{x}{2})$	Period ×2 → 4π (horizontal stretch)

Rule: $a \cdot f(bx)$. a = vertical scale. b = horizontal speed ($b > 1$ squeezes, $b < 1$ stretches).

Graph 8: Absolute Value & Scaling Transformations



Graph 9A: Stretch and Shrink — $a \cdot f(bx)$



Graph 9A: Left — Vertical stretch changes amplitude. Right — Horizontal stretch changes period. In $a \cdot f(bx)$, a controls height, b controls width.

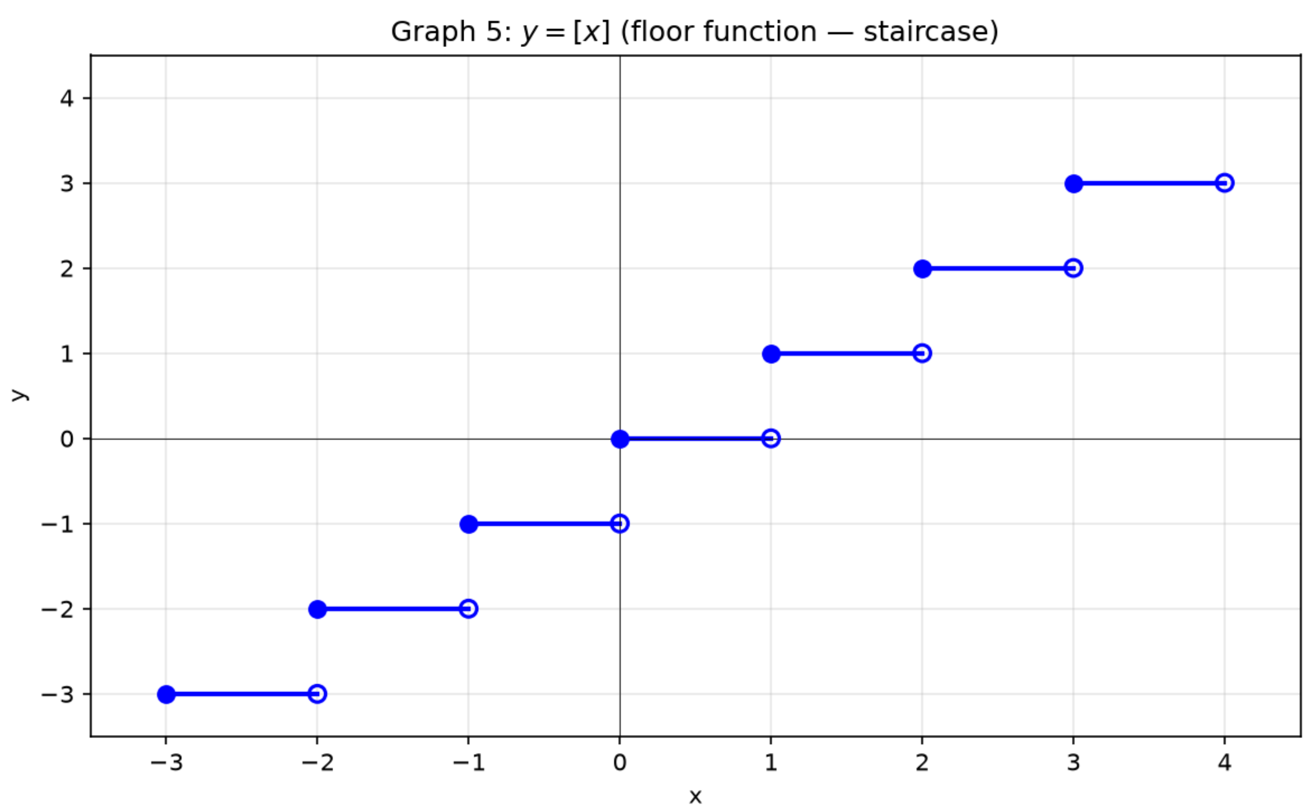
Part C: Special Functions — Stairs, Sawteeth, and Signs

Example 11: Floor Function $\lfloor x \rfloor$ — Building Stairs

$\lfloor x \rfloor$ = greatest integer $\leq x$.

Values: $\lfloor 0.3 \rfloor = 0$, $\lfloor 0.999 \rfloor = 0$, $\lfloor 1.7 \rfloor = 1$, $\lfloor -0.3 \rfloor = -1$.

How to draw: On $[0, 1)$: horizontal at $y = 0$, right endpoint empty. On $[1, 2)$: horizontal at $y = 1$, left endpoint filled, right empty. Repeat left and right forever.

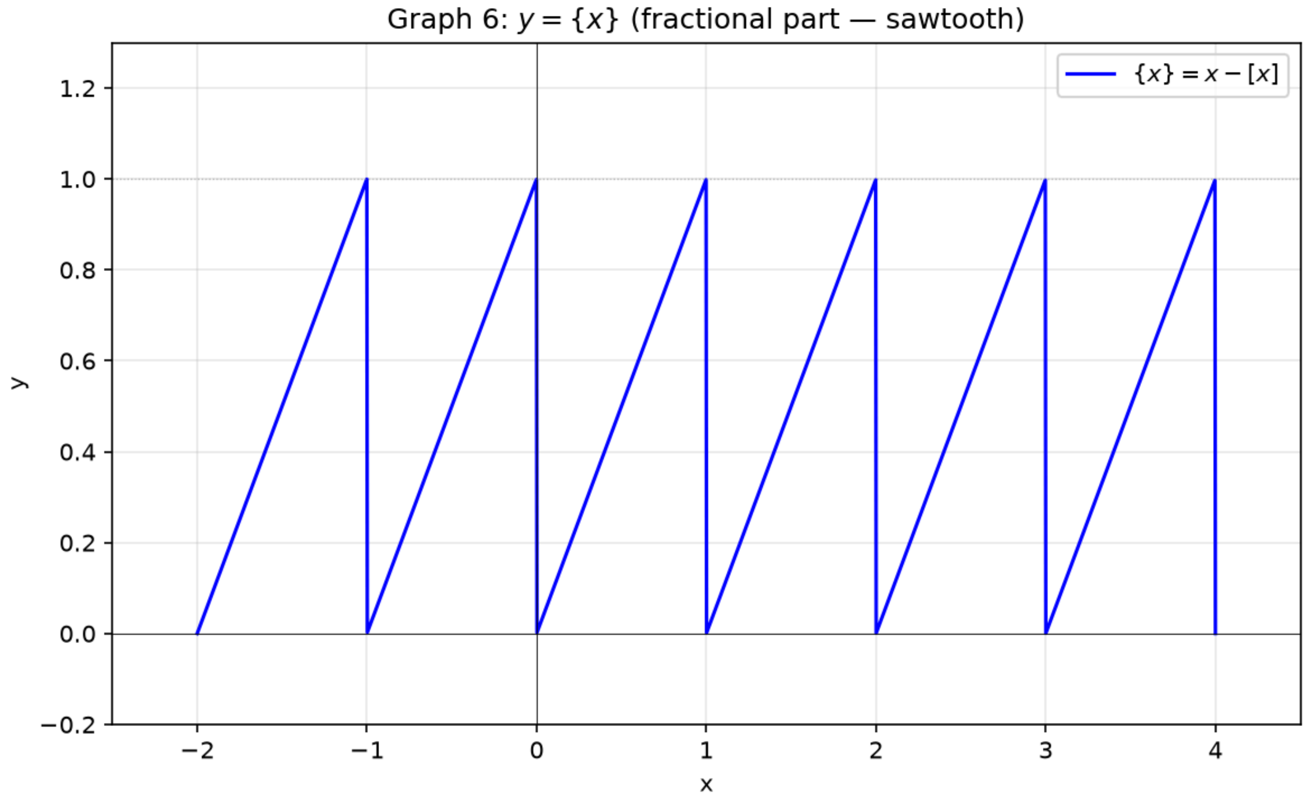


Example 12: Fractional Part $\{x\}$ — Repeating Sawteeth

$\{x\} = x - \lfloor x \rfloor$ = the decimal part only.

$\{3.7\} = 0.7$, $\{5.0\} = 0$, $\{-1.2\} = -1.2 - (-2) = 0.8$.

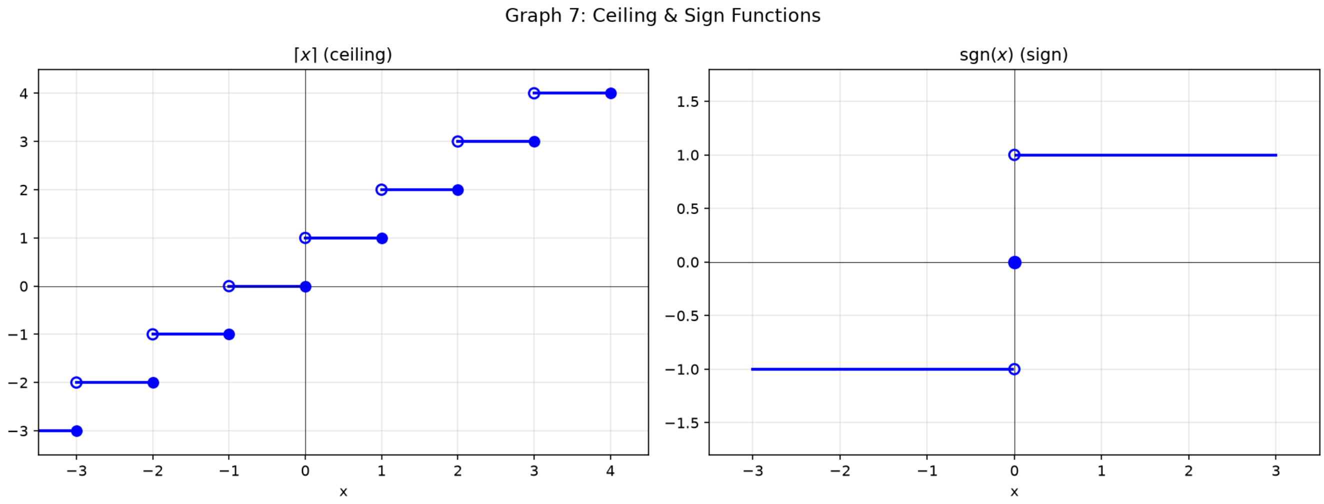
How to draw: On $[0, 1)$: $\{x\} = x$, diagonal from $(0, 0)$ to just before $(1, 1)$. On each $[n, n + 1)$: same diagonal shifted. Height always in $[0, 1)$.



Example 13: Ceiling and Sign Functions

$\lceil x \rceil$: smallest integer $\geq x$. $\lceil 3.2 \rceil = 4$, $\lceil -1.2 \rceil = -1$.
Like floor but shifted right: $(0, 1] \to 1$, $(1, 2] \to 2$.

$\text{sgn}(x)$: $x < 0 \to -1$, $x = 0 \to 0$, $x > 0 \to 1$. Three horizontal segments.



Example 14: Piecewise — Draw Each Interval Separately

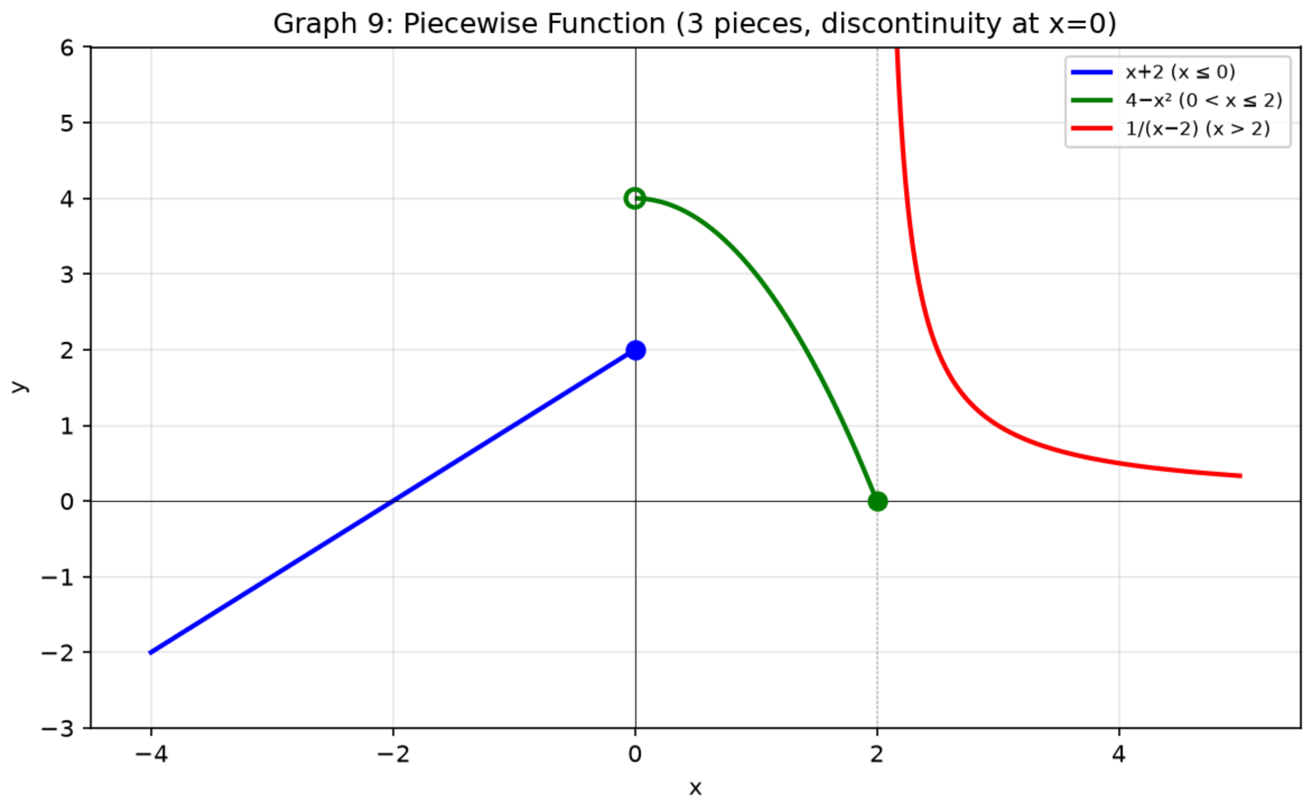
$$f(x) = \begin{cases} x + 2, & x \leq 0 \\ 4 - x^2, & 0 < x \leq 2 \\ \frac{1}{x-2}, & x > 2 \end{cases}$$

Check boundaries first:

$x = 0$: piece 1 gives 2. Piece 2 gives 4. → **Jump!** Filled dot $(0, 2)$, empty $(0, 4)$.

$x = 2$: piece 2 ends at 0. Piece 3 denominator → 0. → Vertical asymptote at $x = 2$.

Draw: Piece 1: line ending at $(0, 2)$. Piece 2: parabola from $(0, 4)$ to $(2, 0)$. Piece 3: hyperbola with wall at $x = 2$, approaching $y = 0$.

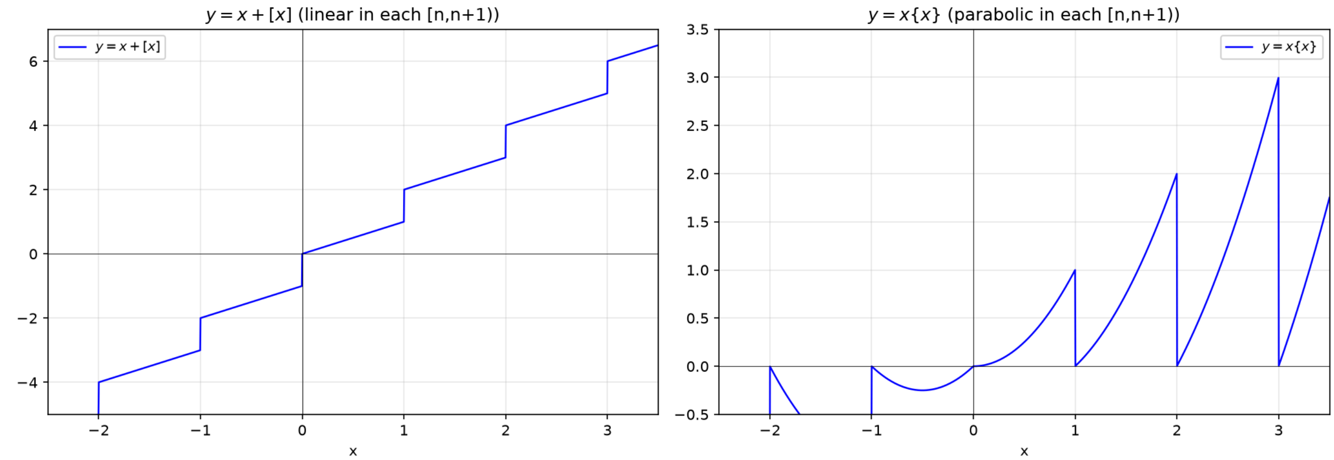


Example 15: Mixed — Floor × Fractional Part

$f(x) = x + \lfloor x \rfloor$. Write $x = n + \delta$ (n integer, $0 \leq \delta < 1$).
$f(x) = n + \delta + n = 2n + \delta$. On $[0, 1)$: $y = \delta$ (diagonal 0→1). On $[1, 2)$: $y = 2 + \delta$ (diagonal 2→3). Jumps by 1 at integers.

$f(x) = x\{x\} = x(x - \lfloor x \rfloor)$. On $[0, 1)$: x^2 (parabola piece). On $[1, 2)$: $x(x - 1)$ (another parabola piece). Drops to 0 at every integer.

Graph 10: Mixed Floor/Fractional Part Functions



Example 16: Absolute Value + Rational

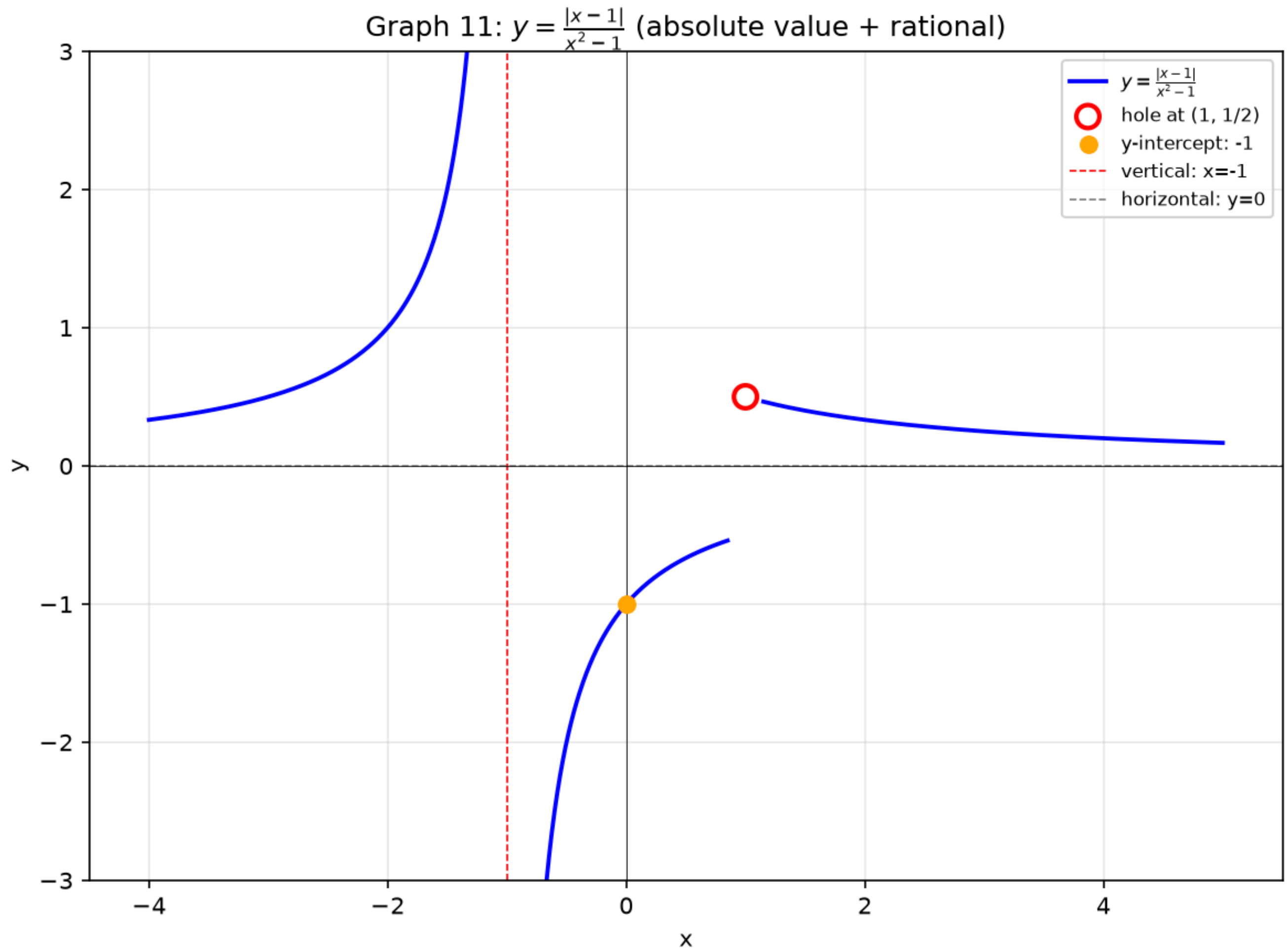
$f(x) = \frac{|x-1|}{x^2-1}$.

Strip abs: $x \geq 1$: $|x-1| = x-1$, $f(x) = \frac{1}{x+1}$ $(x \neq 1)$.
$x < 1$: $|x-1| = -(x-1)$, $f(x) = \frac{-1}{x+1}$ $(x \neq -1)$.

Hole: at $x = 1$, value would be $\frac{1}{2}$ → empty circle $(1, \frac{1}{2})$.

Asymptotes: vertical $x = -1$, horizontal $y = 0$.

Draw: Two branches: $x < -1$ (upper right quadrant), $-1 < x < 1$ (lower), $x > 1$ (upper right approaching 0).



> **Up to here**: Transform = move/flip/fold/stretch. Floor = stairs. Frac = sawteeth. Piecewise = interval-by-interval. Mixed = combine patterns.

Common Mistakes

Mistake 1: Forgetting the hole after canceling

Wrong: "$\frac{(x-1)(x+2)}{x-1} = x + 2$, so it's a line." **Right**: Punch an empty circle at $x = 1$. The original denominator forbids it forever.

Mistake 2: Filling the right endpoint of a floor step

Wrong: On $[1, 2]$ the value is 1. **Right**: $\lfloor 2.0 \rfloor = 2$. The interval is $[1, 2)$ — right endpoint empty.

Mistake 3: Assuming a graph never crosses its horizontal asymptote

Wrong: "Asymptote means forbidden." **Right**: Asymptote only describes $x \to \pm\infty$ behavior. The graph can cross it at finite x. Example: $y = \frac{x}{x^2+1}$ has $y = 0$ asymptote but passes through $(0, 0)$.

Mistake 4: $f(|x|)$ vs $|f(x)|$ confusion

Wrong: They're the same. **Right**: $|f(x)|$ folds below-axis parts upward. $f(|x|)$ copies right side to left, making an even function.

What We Just Did

```
(1) 7-step drawing sequence: domain → symmetry → intercepts → asymptotes
    → sign → connect → special handling.

(2) Transformations: move (±h,±k), flip (-f, f(-x)),
    fold (|f|, f(|x|)), stretch (a·f(bx)).

(3) Special functions: [x] (stairs), {x} (sawteeth), ⌈x⌉ (ceiling), sgn(x).
    Piecewise: check boundaries, draw interval by interval.
```

Practice 1

Find the domain of $f(x) = \frac{\sqrt{9-x^2}}{\ln(x-1)}$. State all conditions.

→ Reference: **Example 1**

> Solutions: Solutions

Practice 2

Draw $f(x) = \frac{x^2-4}{x^2-1}$ using the 7 steps. Show asymptotes, intercepts, sign, and any holes.

→ Reference: **Example 3**

> Solutions: Solutions

Practice 3

Draw $f(x) = \frac{x^2+1}{x-2}$. Find the slant asymptote by dividing.

→ Reference: **Example 4**

> Solutions: Solutions

Practice 4

The graph of $f(x) = \sqrt{x}$ is shifted left 3 and down 1. Write the equation and state the domain.

→ Reference: **Example 6, 7**

> Solutions: Solutions

Practice 5

Draw $f(x) = |x^2 - 4|$. Identify which part was folded upward.

→ Reference: **Example 9**

> Solutions: Solutions

Practice 6: Real Battle

$$f(x) = \begin{cases} -x-2, & x < -1 \\ x^2, & -1 \le x < 2 \\ \frac{4}{x-1}, & x \ge 2 \end{cases}$$

Draw the graph. Mark all boundary dots as filled or empty. Check continuity at each boundary.

→ Reference: **Example 14**

> Solutions: Solutions

Basic Algebra Drill — Graph Drawing (10 Problems)

> Pure computation. Domain, asymptotes, intercepts.

D1. Find the domain of $f(x) = \sqrt{2x+6}$.

D2. Find the domain of $f(x) = \frac{1}{x^2-9}$.

D3. Find all x- and y-intercepts of $f(x) = x^3 - 9x$.

D4. Find all vertical and horizontal asymptotes of $f(x) = \frac{3x}{x-2}$.

D5. Find the slant asymptote of $f(x) = \frac{x^2+3x}{x+1}$.

D6. Write the equation after shifting $y = |x|$ right 4 and up 2.

D7. Write the equation after reflecting $y = \sqrt{x}$ across the y-axis and shifting up 3.

D8. Compute $\lfloor 3.7 \rfloor$, $\lfloor -2.3 \rfloor$, $\lceil 1.2 \rceil$, $\lceil -0.8 \rceil$.

D9. For $f(x) = \frac{x^2-1}{x^2-4}$, find where the graph crosses its horizontal asymptote.

D10. A rational function has vertical asymptotes at $x = -1$ and $x = 2$, and a horizontal asymptote at $y = 0$. Propose a possible formula.

> Solutions: Solutions

Advanced Algebra Drill — Graph Drawing (10 Problems)

> Multi-step. Draw, then reason.

A1. $f(x) = \frac{x^3-8}{x-2}$. Cancel, find the hole, and draw the graph. What simple function does it resemble?

A2. $f(x) = \frac{|x-2|}{x-2}$. Determine the formula on $x > 2$ and $x < 2$. Draw. What is the range?

A3. Draw $f(x) = \lfloor 2x \rfloor$ on $[-2, 2]$. How does the stair width compare to $\lfloor x \rfloor$?

A4. $f(x) = \frac{x^2-5x+6}{x^2-x-6}$. Factor numerator and denominator. Find holes (if any) and asymptotes.

A5. $f(x) = \frac{1}{(x-1)^2}$. State domain, asymptotes, sign, and range. Draw.

A6. $f(x) = x + \frac{1}{x}$. Find the slant asymptote. Show there are no vertical asymptotes at $x = 0$ — what happens instead?

A7. Draw $f(x) = \{x\} - \frac{1}{2}$ on $[-2, 3]$. Describe how the sawtooth is shifted.

A8. $f(x) = \sqrt{x^2 - 4}$. Find the domain. Show that as $x \to \infty$, the graph approaches the line $y = x$. Draw.

A9. Draw $f(x) = ||x| - 2|$. Build from the inside out: $|x| \to |x| - 2 \to ||x| - 2|$. How many V-shaped folds?

A10. Propose a piecewise formula whose graph has: a parabola on $(-\infty, 0]$, a constant from $(0, 2]$, and an exponential decay for $x > 2$, with a hole at $x = 0$ and a jump at $x = 2$.

> Solutions: Solutions

Today's Procedure

```
Step 1: 7-step sequence — master it. Domain (4 rules), symmetry test,
         intercepts (set x=0, set y=0), asymptotes (vertical/horizontal/slant),
         sign chart, connect dots, handle holes/jumps.

Step 2: Transformations — any graph = base shape + move + flip + fold + stretch.
         Piecewise = interval-by-interval with boundary checks.

Step 3: Special functions — [x] staircase, {x} sawtooth. They repeat.
         Mix them with algebra to create new patterns.
```

Terminology

What we call it	Math term	Notation
numbers you can push in	domain	$\{x \mid \text{conditions}\}$
numbers that come out	range	$\{f(x) \mid x \in \text{domain}\}$
moving the graph	translation	$f(x-h)+k$
flipping over axis	reflection	$-f(x), f(-x)$
folding upward	absolute value of function	$\lvert f(x) \rvert$
stretching/shrinking	scaling	$a \cdot f(bx)$
stair function	floor / greatest integer	$\lfloor x \rfloor$
ceiling function	ceiling	$\lceil x \rceil$
sawtooth function	fractional part	$\{x\} = x - \lfloor x \rfloor$
sign function	signum	$\operatorname{sgn}(x)$
piece-by-piece	piecewise function	different formulas per interval

What we call it	Math term	Notation
hole	removable discontinuity	canceled factor
line it hugs	asymptote	vertical, horizontal, slant